TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Observation of the Double-Slit Interference Pattern

Young's fringes observed and measured — equal spacing β = λD/d, the central maximum, and what changes when the colour, the slit gap or the screen distance changes.

PHY-1104 · PHYSICS-I SESSIONAL · EXPERIMENT 16 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to observe the interference pattern of a double slit, verify its properties, and measure the fringe width. Light from a narrow source slit illuminates two parallel slits separated by d. The two slits act as coherent sources (they are fed by the same wavefront — the coherence condition of Physics-I Topic 6), and on a screen at distance D their waves reinforce and cancel in a row of equally spaced bright and dark bands. The fringe width — the spacing between consecutive bright (or dark) fringes — is

$$\beta = \lambda D/d$$

The central fringe (zero path difference) is the brightest and, with white light, the only purely white one: at every other position the colours' fringe systems, having different β, fall out of step and tint the bands.

2. Apparatus and procedure

- **Apparatus:** optical bench, sodium lamp (or laser pointer where the laboratory uses one) with a single slit, a double-slit slide of known separation, screen with scale (or travelling microscope), white-light source for the colour observation.
- Set the source slit, double slit and screen coaxial; obtain a clear row of vertical fringes in a darkened room.
- **Observe and record:** the equality of spacing; the central bright fringe; the gradual fall of brightness outward; the effect of covering one slit (fringes vanish into a single-slit glare); the effect of white light (white centre, coloured edges, pattern washing out after a few fringes).
- **Measure:** with the scale/microscope, the distance spanning 10 consecutive fringes; β is one-tenth of it. Repeat at another D, and (if slides allow) another d, checking $\beta \propto D$ and $\beta \propto 1/d$.
- Compute λ from β, d and D and compare with the source's known value.

3. Worked calculation (model data)

RESULT

Double slit: d = 0.50 mm = 5.0×10^{-4} m. Screen distance D = 1.00 m. Sodium light.
Ten fringes span 11.8 mm → β = 1.18 mm = 1.18×10^{-3} m.
$\lambda = \beta d/D = (1.18 \times 10^{-3} \times 5.0 \times 10^{-4})/1.00 = 5.90 \times 10^{-7}$ m = **590 nm** — sodium's 589.3 nm, recovered from a ruler measurement of light bands. Doubling D to 2.00 m in the repeat run doubled the span to 23.6 mm, verifying $\beta \propto D$ directly.

4. Observations to record (the experiment's real content)

CHANGE MADE	WHAT IS OBSERVED, AND THE REASON
One slit covered	Fringes disappear; a broad single-slit diffraction glare remains — interference needs two coherent beams.
White light replaces sodium	Central fringe white; fringes beside it coloured (violet edge inward, red outward — β grows with λ); pattern washes out within a few orders as the colours' systems dephase.
Screen moved farther (D ↑)	Fringes widen in proportion ($\beta \propto D$) and grow dimmer.
Wider slit separation (d ↑)	Fringes narrow ($\beta \propto 1/d$).
Red filter vs blue filter	Red fringes wider than blue ($\beta \propto \lambda$), a direct visual ranking of wavelengths.

5. Precautions

- The source slit must be narrow and the two slits evenly illuminated, or the fringes are faint and unequal.
- Measure across many fringes (10 or more) and divide — a single β is smaller than the reading error of its edges.
- Keep the screen perpendicular to the axis and the room dark; stray light washes out the dark fringes first.
- Measure D from the double slit's plane to the screen's surface, not to the bench uprights' indices without correction.
- If a laser is used, never look into the beam or its specular reflections; view only the screen.

6. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why do the two slits produce interference while two separate lamps side by side do not?
2. Fringes of width 0.90 mm are formed with D = 0.80 m and d = 0.40 mm. Find λ.
3. Why is the central fringe the brightest, and why does brightness fall off sideways?
4. In white light, which colour's edge of a bright fringe lies nearer the centre, and why?
5. What single observation in this experiment most directly proves light behaves as a wave?
6. The whole apparatus is immersed in water (μ = 1.333). What happens to β?

Answer key

1. Interference needs a fixed phase relation between the two beams. The two slits are illuminated by one wavefront from one source, so their phase difference is locked (coherence). Two independent lamps emit in unrelated bursts; any pattern averages away faster than the eye or detector can follow.
2. $\lambda = \beta d/D = (0.90 \times 10^{-3} \times 4.0 \times 10^{-4})/0.80 = 3.60 \times 10^{-7}/0.80 = 4.50 \times 10^{-7}$ m = 450 nm — blue light.
3. At the centre the path difference is zero for all wavelengths, so every colour reinforces there — the strongest, whitest maximum. Sideways, the fall-off comes from the single-slit diffraction envelope of each slit's own pattern: the pair's interference fringes are modulated by each slit's diffraction, which fades with angle.
4. The violet (shorter-λ) edge lies nearer the centre: $\beta = \lambda D/d$ is smallest for violet, so at any off-centre position the violet system has its bright fringe at the smaller displacement; red, with the largest β, fringes farthest out.
5. The existence of perfectly dark fringes — places where light added to light produces darkness. Particle pictures cannot cancel; only waves with phase can destructively interfere. (The equal spacing and its λ-proportionality then quantify the wave behaviour.)
6. The wavelength in water shortens to λ/μ, so β shrinks by the factor 1.333: $\beta_{water} = \beta_{air}/1.333 = 0.75\,\beta_{air}$ — fringes close up by a quarter, a neat in-situ measure of μ.